

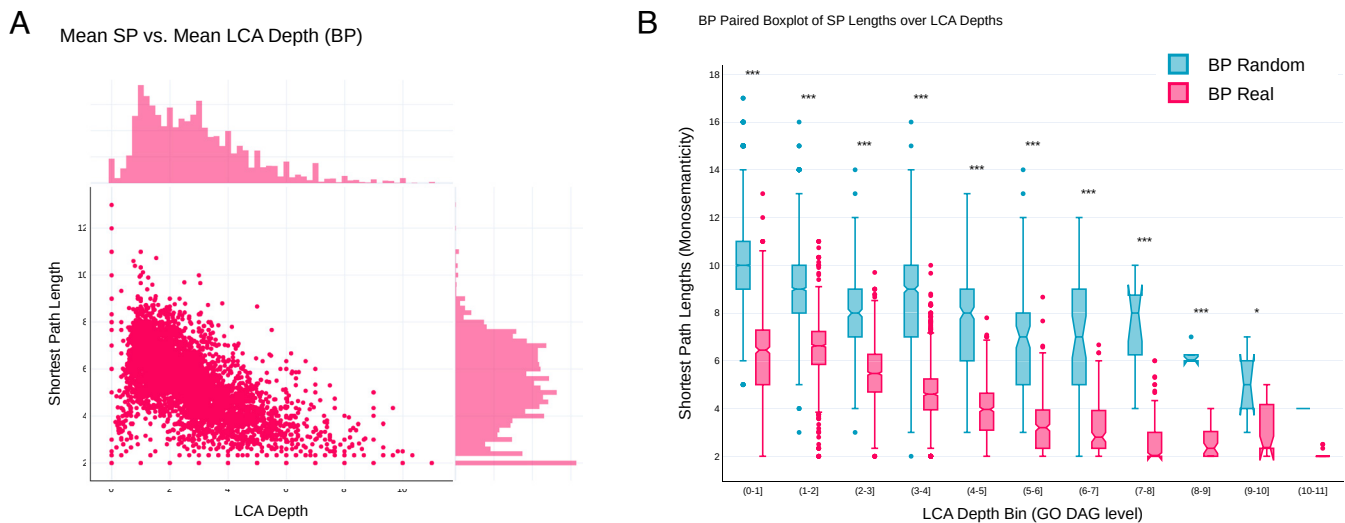


Fig. 2. Evidence of biological process specific neurons. Each neuron (point) is associated with a collection of proteins that it activates. GO Enrichment was conducted for each of these protein sets to associate GO terms per neuron. (A) Scatterplot shows a comparison of the average pairwise shortest path length and least common ancestor among each neuron's enriched GO Terms. Significant number of neurons have low shortest path lengths (monosemantic) across all levels of the GO DAG. (B) Grouped boxplot shows comparison to random pairwise shortest path lengths over each least common ancestor (LCA) depth. All results shown for sparse features from SAE trained on protein-level representations of ESM2 Layer 7.

feature's enriched set of GO terms, we computed each pairwise least common ancestor (LCA). The mean of the depths of these LCAs was calculated to indicate the approximate position or depth (distance from root) of the GO terms enriched by that sparse feature. This metric provided a quantitative understanding of whether the sparse feature was associated with more general (higher-level) or specific (lower-level) functional annotations.

These two metrics—GO monosemanticity and position on the GO DAG—allowed us to characterize individual sparse features. Notably, we observed strong evidence of neurons that are functionally monosemantic (by our definition above) across all depths of the GO hierarchy as indicated by the low shortest path lengths across all LCA depths in Fig. 2A. Examples of these neurons and associated GO Terms are organized in Table 1. To understand the significance of this result, we established a random baseline by computing pairwise shortest path lengths and LCA depths for randomly sampled GO terms that were linked with the specific set of human proteins that we used. At every depth of the GO DAG (LCA Depth), the pairwise shortest path lengths of each neuron's GO terms were significantly smaller than random (Fig. 2B). Analogous figures for cellular component and molecular function respectively are in [SI Appendix, Figs. S1–S4](#).

With this analysis, we demonstrated that certain sparse features could be mapped to the GO DAG across all levels of the hierarchy, from very specific leaf-level nodes to broader functional categories. Notably, this mapping was achieved in an entirely unsupervised manner, leveraging only the information extracted from the ESM protein-level representations, without any direct supervision or prior GO knowledge.

Finally we remark on the extent to which GO terms in the DAG and the protein set used for GO analysis are covered by sparse features (neurons). The total number of enriched GO terms for layer 9's SAE is 9789. These are partitioned into 6024, 2740, and 1025 biological process, molecular function, and cellular component GO terms, resp. In total there are 26467, 10146, and 4022 GO terms in the DAG belonging to the three namespaces biological process, molecular function, and cellular component, resp. Of these, 11545, 4354, 1713 GO terms appear in our set of GO terms used in the protein set used for our GO

analysis; thus, for layer 9's SAE, 52.17%, 62.93%, 59.83% of the GO terms present in the set used for GO analysis feature among the set of GO terms enriched by neurons. [SI Appendix, Table S3](#) provides this information for all other layers and models. We also investigate the degree of overlap between enriched GO terms found by SAEs and transcoders for the same layer. For example, in layer 9's SAE, we find 9789 enriched GO terms whereas for layer 9's transcoder we find 9659 enriched GO terms. Of these, the two share 9180 enriched GO terms in common. Thus, both methods (SAEs and transcoders) have a majority of their enriched GO terms in common. [SI Appendix, Table S4](#) indicates the degree of overlap for all layers.

LLM-Assisted Interpretation Finds Biologically Relevant Sparse Features. Once we have our sparse features from the SAEs and transcoders, we draw from work in natural language processing (9, 11, 19, 20) to automate the interpretation of these sparse features using an LLM; automation is critical due to the large number of sparse features. We do this by selecting a sparse feature (neuron) from among 20,000 that is sufficiently active on a 50,000 sequence subset of UniProt. We pick 17 active and 15 inactive sequences for this sparse feature and as context we provide Claude[‡] with half of these sequences' ID, activation value, protein names, gene names, protein families, and GO terms, including biological processes, cellular components, molecular functions—and asked Claude to come up with a single sentence long human-readable interpretation for what our SAE's sparse feature[§] appears to be capturing. We aimed to assess the interpretability of sparse features in the context of this metadata. So we then score the Claude-provided feature interpretation by first asking Claude (in a separate chat) to use this interpretation to predict (i.e. simulate) the feature's activations using the same metadata for the withheld half of the sequences from above. Then following work in the NLP setting (9, 11, 19), we compute the Pearson correlation between the actual activation values and the simulated ones that Claude predicts, and this is regarded as the interpretability

[‡]Claude 3.5 Sonnet.

[§]The SAE's sparse features are simply neurons in the SAE's hidden sparse layer.

Table 1. Examples of sparse neuron’s associated with specific biological processes across all levels of the GO DAG

Neuron	LCA depth	Terms
2793	1	["viral translational frameshifting (GO:0075523)," "viral process (GO:0016032)," "aspartic-type endopeptidase activity (GO:0004190)," "aspartic-type peptidase activity (GO:0070001)"]
10282	2	["retrotransposition (GO:0032197)," "transposition (GO:0032196)"]
9095	3	["peptide modification (GO:0031179)," "peptide metabolic process (GO:0006518)"]
16984	4	["positive regulation of viral budding via host ESCRT complex (GO:1903774)," "regulation of viral budding via host ESCRT complex (GO:1903772)"]
7631	5	["metallo-sulfur cluster assembly (GO:0031163)," "iron-sulfur cluster assembly (GO:0016226)"]
953	6	["protein maturation by [4Fe-4S] cluster transfer (GO:0106035)," "protein maturation by iron-sulfur cluster transfer (GO:0097428)"]
1105	7	["iron ion transport (GO:0006826)," "transition metal ion transport (GO:0000041)"]
8646	8	["positive regulation of protein oxidation (GO:1904808)," "regulation of protein oxidation (GO:1904806)"]
3823	9	["leukotriene D4 catabolic process (GO:1901749)," "leukotriene catabolic process (GO:0036100)," "dipeptidase activity (GO:0016805)," "metallo-dipeptidase activity (GO:0070573)"]
11233	10	["positive regulation of serine-type endopeptidase activity (GO:1900005)," "positive regulation of serine-type peptidase activity (GO:1902573)"]
3771	11	["pentose-phosphate shunt, oxidative branch (GO:0009051)," "NADPH regeneration (GO:0006740)," "glucose-6-phosphate dehydrogenase activity (GO:0004345)"]

Only sparse neurons whose average shortest path length among its enriched GO Terms was ≤ 2 were chosen. One sparse neuron per least common ancestor (LCA) Depth (GO DAG Level) was selected, and the higher the LCA Depth, the more niche the biological process. Table was created using sparse neurons (features) from the SAE trained on the protein-level representation of ESM2 Layer 7.

score for the feature. Using Claude for this enables a scalable method of scoring a feature’s interpretability. We employ this autointerpretability method for sparse features from all our SAEs and transcoders (See *Materials and Methods* for more details.)

SAE results for protein-level representations. We demonstrate that many sparse features extracted from the SAEs trained on protein-level representations are very interpretable, as evidenced by the Pearson correlation scores in Table 2. Across all examined layers, the median Pearson correlations are near or above 0.7. Table 3 presents examples of Claude’s automated interpretations for both highly interpretable (0.9+ scoring) and near-median level interpretable sparse features. For instance, a sparse feature

(2579 in layer 7) is activated by proteins with methyltransferase activity, especially in RNA and DNA methylation. Additionally, certain sparse features (11982 in layer 9, 13811 in layer 7, and 19647 in layer 8) are linked to specific protein families and their roles, such as the NAD kinase family (NAD metabolic processes and NADP biosynthesis), Diaminopimelate epimerase family (lysine biosynthesis via diaminopimelate with diaminopimelate epimerase activity) and the IUNH family (nucleoside hydrolase activity). Another sparse feature (16397 in layer 10) is associated with proteins involved in pseudouridine synthesis, especially tRNA and rRNA pseudouridine synthesis, from the TruB and RluA families, while sparse feature 15683 in layer 10 is linked

Table 2. Comparison of sparse autoencoder (SAE), Transcoder (TC), and ESM neurons autointerpretability scores for various layers for the protein-level representations

Model (Layer)	Median correlation	Mean correlation	% with $P < 0.05$	Neurons analyzed	Neurons attempted
Sparse (layer 6)	0.706	0.634	76%	200	819/20,000
TC (layer 6)	0.710	0.635	78.5%	200	882/20,000
ESM (layer 6)	0.52	0.443	52.8%	200	202/480
Sparse (layer 7)	0.702	0.629	72%	200	703/20,000
TC (layer 7)	0.734	0.640	75.5%	200	598/20,000
ESM (layer 7)	0.471	0.418	48.2%	200	201/480
Sparse (layer 8)	0.726	0.623	73%	200	585/20,000
TC (layer 8)	0.704	0.649	77.5%	200	575/20,000
ESM (layer 8)	0.454	0.42	49.5%	200	202/480
Sparse (layer 9)	0.692	0.618	69%	200	684/20,000
TC (layer 9)	0.682	0.638	76%	200	565/20,000
ESM (layer 9)	0.425	0.398	43%	200	203/480
Sparse (layer 10)	0.756	0.659	76%	200	601/20,000
TC (layer 10)	0.667	0.601	69%	200	629/20,000
ESM (layer 10)	0.435	0.383	42%	200	201/480

Sparse and TC neurons are more interpretable than ESM neurons. (Note that for Sparse and TC there are 20,000 neurons to pick from to analyze and for ESM there are 480 neurons to pick from, but we only analyze a neuron’s interpretability if it is sufficiently active. This is why there are more neurons attempted than analyzed especially for Sparse and TC.) Here, P -value means P -value of the Pearson correlation (i.e. the autointerpretability score), so that column indicates the percentage of neurons with $P < 0.05$.

Table 3. Examples of several SAE neurons (from various layers) with very high interpretability and some median interpretability, with Claude’s interpretation, for protein-level representations

Model	Neuron no.	Claude’s interpretation	Correlation
Sparse (layer 9)	6384	This neuron appears to be primarily detecting proteins involved in phosphoenolpyruvate carboxykinase (ATP) activity and gluconeogenesis, with a strong focus on the Phosphoenolpyruvate carboxykinase (ATP) family.	0.972
Sparse (layer 9)	11982	This neuron appears to be detecting proteins that belong to the NAD kinase family and are involved in NAD metabolic processes and NADP biosynthesis.	0.954
Sparse (layer 7)	13811	This neuron appears to be detecting proteins belonging to the Diaminopimelate epimerase family, specifically those involved in lysine biosynthesis via diaminopimelate with diaminopimelate epimerase activity.	0.993
Sparse (layer 7)	2579	This neuron appears to be detecting proteins with methyltransferase activity, particularly those involved in RNA or DNA methylation processes.	0.930
Sparse (layer 8)	19647	This neuron appears to be detecting proteins with nucleoside hydrolase activity, particularly those belonging to the IUNH family and involved in nucleobase metabolic processes.	0.999
Sparse (layer 8)	3800	This neuron appears to be detecting proteins involved in transmembrane transport of ions or amino acids, particularly those located in the plasma membrane.	0.721
Sparse (layer 10)	16397	This neuron appears to be detecting proteins involved in pseudouridine synthesis, particularly tRNA and rRNA pseudouridine synthases from the TruB and RluA families.	0.969
Sparse (layer 10)	15683	This neuron appears to be detecting proteins involved in the electron transport chain, particularly NADH dehydrogenase and quinone oxidoreductase components of Complex I.	0.758

The correlation score, ranging between 0 and 1, represents the interpretability of the SAE neuron. See [SI Appendix, section S5](#) for full list.

to Complex I proteins of the electron transport chain. These findings describe the ability of SAEs to disentangle meaningful biological signals from protein-level representations in an entirely unsupervised manner. This enhances the interpretability of these representations that frequently get used in downstream protein-level tasks.

We compare the automated interpretability scores of sparse features (Table 2) with those of ESM neurons in the protein-level representation (see *Materials and Methods* for details). Across the examined layers, ESM neurons exhibit median Pearson correlations slightly above 0.5 at best, whereas sparse features consistently achieve median correlations around 0.7. This demonstrates that the SAE-derived features are more interpretable than ESM neurons. Fig. 3A provides a visual comparison, showing an overlaid histogram of interpretability scores for sparse features extracted from the SAE trained on layer 9, alongside ESM neuron scores from the same layer. The distribution highlights a notable number of highly interpretable sparse features (with Pearson correlations between 0.7 and 1), compared to a smaller number for ESM neurons. This reinforces the advantage of SAEs in disentangling biologically meaningful information from protein-level representations.

SAE results for amino acid-level representations. We also aim to functionally interpret SAE features extracted from SAEs trained on amino acid (AA) level representations. For this we follow the same autointerpretability method as we do for SAE features from the SAE trained on protein-level representations—however, since protein sequences can have a large number of amino acids and this number can be very different for different protein sequences, to sustain feasibility as in the protein-level case, for an SAE feature (or neuron in the case of ESM) we aim to interpret the mean activation of that SAE feature (or neuron) across the entire protein sequence.

We show that many sparse features from the SAEs trained on AA-level representations are also very interpretable, as

demonstrated by the interpretability scores (Pearson correlations) in Table 5. The median Pearson correlations are 0.673 to 0.714 across the layers examined. Table 4 highlights several highly interpretable (around 0.9 scoring) and some near median-level interpretable sparse features extracted by SAEs trained on amino acid-level representations, demonstrating their ability to capture functionally meaningful patterns. Many of these features align closely with known protein families and functions, and at least one with an uncharacterized protein family. For example, sparse feature 19717 (layer 9) is associated with proteins from the UPF0312 family (uncharacterized), while feature 7554 (layer 10) is associated with proteins involved in olfactory and gustatory sensory perception. Certain sparse features correspond to specific enzyme families like feature 15537 (layer 9), linked to the PTH family with peptidyl-tRNA hydrolase activity and involved in translation processes, and feature 12791 (layer 7) associated with the Metallo-dependent hydrolases superfamily and Adenine deaminase family. Another sparse feature (10650 in layer 7) is linked to proteins with calcium ion binding functionality. These findings indicate that SAEs can (in an unsupervised manner) extract functionally relevant signals from amino acid-level representations, revealing several functionally distinct protein families and metabolic pathways.

We compare the automated interpretability score of sparse features in Table 5 with that of ESM neurons in the AA-level representation (see *Materials and Methods* for details). The median Pearson correlations for ESM neurons peaks just below 0.5 across examined layers, whereas SAE features achieve a median score of 0.673 to 0.714, demonstrating their superior interpretability for AA-level representations as well—and comparable to the performance for protein-level SAEs. Fig. 3C presents an overlaid histogram with automated interpretability scores of layer 9’s SAE and layer 9’s ESM neurons for the AA-level representations.

Transcoder results for protein-level representations. Transcoders (14) were originally developed to create a sparse approximation